Fiscal Year:	FY 2004	**Task Last Updated:**	FY 04/03/2006
PI Name:	Smith, C. Wayne M.D.		
Project Title:	Effects of Hind Limb Suspension on Local Inflammation and Wound Healing in a Murine Model		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Physiology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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PI Organization Type:	UNIVERSITY	**Phone:**	713-798-0310
Organization Name:	Baylor College of Medicine		
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PI Address 2:	1100 Bates, Room 6014		
PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77030-2600	**Congressional District:**	25
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	99-HEDS-03
Start Date:	09/01/2000	**End Date:**	09/30/2004
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	0	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	None		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	A specific aim of this grant was to assess the effects of hind limb unloading (HU) on wound healing. Our previous studies have shown that HU induces liver injury as a result of increased portal endotoxemia, and the current work extends these studies to the evaluation of the inflammatory response associated with wound healing. We have chosen as a model superficial abrasion of the corneal epithelium. This model allows accurate documentation of re-epithelialization and the attendant inflammatory response, including neutrophil, dendritic cell, and Langerhan cell mobilization. Our results reveal a markedly delayed healing and a markedly reduced inflammatory response.		
Rationale for HRP Directed Research:			

Research Impact/Earth Benefits:	The overall objective of this work is to determine a model of altered inflammation and wound healing, reflecting the possible conditions experienced by individuals in zero gravity. The significance of such a model would be to allow definition of possible molecular and cellular mechanisms, and to serve as a basis for development of therapies to promote effective inflammation and wound healing.
Task Progress:	No final report received. See Bibliography for any publications.
Bibliography Type:	Description: (Last Updated: 01/25/2021)
Articles in Peer-reviewed Journals	Rivera CA, Tcharmtchi MH, Mendoza L, Smith CW. "Endotoxemia and hepatic injury in a rodent model of hindlimb unloading." J Appl Physiol (1985). 2003 Oct;95(4):1656-63. https://doi.org/10.1152/japplphysiol.00302.2003 ; PMID: 12794033 , Oct-2003